

Regional Hybrid Stacking with XGBoost and Deep Attention for One-Day Soil Moisture Forecasting

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Abstract

This study presents a regional, profile-correlated approach for one-day-ahead soil moisture forecasting that combines gradient-boosted trees (XGBoost) with deep sequential models. By clustering hydro-meteorologically homogeneous stations and emphasizing a station-distribution strategy, we reduce spatial noise and substantially improve forecast skill compared to persistence baselines. The XGBoost-based hybrid stacks achieved high one-day accuracy, demonstrating the practical value of tree-based tabular learning when fused with sequential feature extractors for precision irrigation decision-making.

Keywords: soil moisture forecasting; XGBoost; Transformer; LSTM; regional modeling; station clustering; one-day lead

1 Introduction

Soil moisture is a fundamental state variable in the global hydrological cycle, serving as a critical link between the land surface and the atmosphere [1, 2]. Accurate quantification of moisture content is essential for understanding water and energy exchanges, with timely knowledge forming the foundation for scientific irrigation planning, drought early warning systems, and agricultural productivity management [3–5]. In modern precision agriculture, a primary objective is providing reliable short-term forecasts—specifically with a one-day lead time—to determine whether irrigation is required for the following day [3, 4].

Despite its importance, soil moisture dynamics are highly complex, characterized by vertical heterogeneity and non-linear responses to atmospheric forcing [4, 7]. While Root Zone Soil Moisture (RZSM) directly dictates the water available for vegetation uptake, it remains challenging to monitor due to the depth-limited penetration of microwave satellite signals and the high cost of in situ sensor networks [1, 4, 6]. However, recent observations indicate a high degree of statistical coupling between moisture levels at various depths [7, 8]. In this study, we leverage the strong vertical correlation between the surface layer (Volumetric Water Content at 5 cm) and the deeper root zone (up to 100 cm). This profile-correlated approach allows the surface-level dynamics to serve as a robust proxy for the entire soil moisture profile, enabling efficient characterization of the moisture status of the root zone [7, 8].

Traditionally, soil moisture states have been estimated using process-based hydrological models that solve flow and transport equations [4, 9]. However, these models require intensive calibration of soil hydraulic parameters, which are often unavailable or incomplete in data-poor regions [4, 10]. To overcome these limitations, data-driven machine learning (ML) techniques have emerged as a robust alternative, capable of capturing complex non-linear dependencies without prior physical assumptions [4, 11]. Recent advancements have seen a transition from ensemble methods like Random Forest toward deep learning architectures, such as Long Short-Term Memory (LSTM) networks and self-attention Transformer models [3, 12]. Transformers,

in particular, improve predictive stability by dynamically focusing on critical historical time steps, which is vital for variables with high temporal memory [3, 13].

A significant hurdle in national-scale modeling is the inherent spatial heterogeneity across different climatic regimes. Global models trained on diverse datasets often fail to outperform simple persistence baselines, where predictions merely reflect the previous day’s state [4, 14]. Furthermore, while explainable AI frameworks like SHAP (Shapley Additive Explanations) have been used to identify dominant drivers, their effectiveness can be dampened by spatial noise in large, heterogeneous datasets [15, 16]. To address this, we propose a modeling strategy based on **Regional Homogeneous Modeling**. By clustering weather stations with statistically similar hydro-meteorological characteristics, we aim to reduce spatial noise and enhance the signal-to-noise ratio in moisture estimation [17, 18].

This study utilizes a high-resolution dataset from various U.S. weather stations, incorporating primary atmospheric forcing variables: Mean Temperature (*T_DAILY_MEAN*), Solar Radiation (*SOLARAD_DAILY*), and Precipitation (*P_DAILY_CALC*), alongside atmospheric evaporative demand indicators such as Relative Humidity (*RH_DAILY_MAX, MIN, AVG*) [5, 19]. We investigate the performance of advanced standalone and hybrid architectures—including XGBoost, LSTM, CNN, and Transformers—on the primary target of 5 cm Volumetric Water Content. Specifically, we evaluate ”stacked” ensemble models to determine their accuracy in predicting a one-day lag, facilitating data-driven irrigation decision-making [7, 16]. The objectives of this research are to: (i) identify the limitations of global ML models on heterogeneous datasets, (ii) evaluate the efficacy of grouping stations by statistical similarity, and (iii) evaluate the accuracy of hybrid stacking architectures in predicting tomorrow’s moisture levels based on correlated multi-depth profile data.

2 Methodology

2.1 Data Acquisition and Automated Scrapping

The primary data source for this research is the **U.S. Climate Reference Network (USCRN)** provided by the National Oceanic and Atmospheric Administration (NOAA) [7]. To ensure a comprehensive dataset covering diverse climatic regions, an automated Python-based scraping script was developed to extract high-resolution daily soil anomaly data from stations across all U.S. states [7]. The initial dataset contained a broad spectrum of meteorological and soil-state variables, providing a high-dimensional view of land-atmosphere interactions.

2.2 Data Pre-processing and Feature Selection

The raw dataset required extensive cleaning to ensure model stability and predictive accuracy. The first step involved the removal of missing data, specifically identified by the placeholder value of -9999 [18]. The data format was standardized by converting space-separated values into a comma-separated format (CSV), and a *state* column was appended to each row to facilitate regional tracking.

2.2.1 Feature Reduction and Correlation Analysis

To optimize the computational efficiency and avoid the ”curse of dimensionality,” the feature set was reduced from 23 initial variables to a refined set of 12 [7]. This selection was guided by a correlation analysis using the **Pearson Correlation Coefficient** (r), which measures the linear relationship between two variables X and Y as:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \quad (1)$$

Our analysis revealed significant multicollinearity among several features. Specifically, *Mean Temperature* (T_DAILY_MEAN) was found to have a $> 90\%$ correlation with the daily maximum and minimum temperatures, leading to the removal of the latter two to prevent redundancy [7]. Furthermore, a high degree of statistical coupling was observed between the target variable, *Volumetric Water Content at 5 cm* ($SOIL_MOISTURE_5_DAILY$), and deeper soil layers (10 cm, 20 cm, 50 cm, and 100 cm) [7,8]. This vertical correlation (each layer having the highest correlation with its immediate neighbor) confirms that the 5 cm layer serves as a robust proxy for the entire soil moisture profile [7,8].

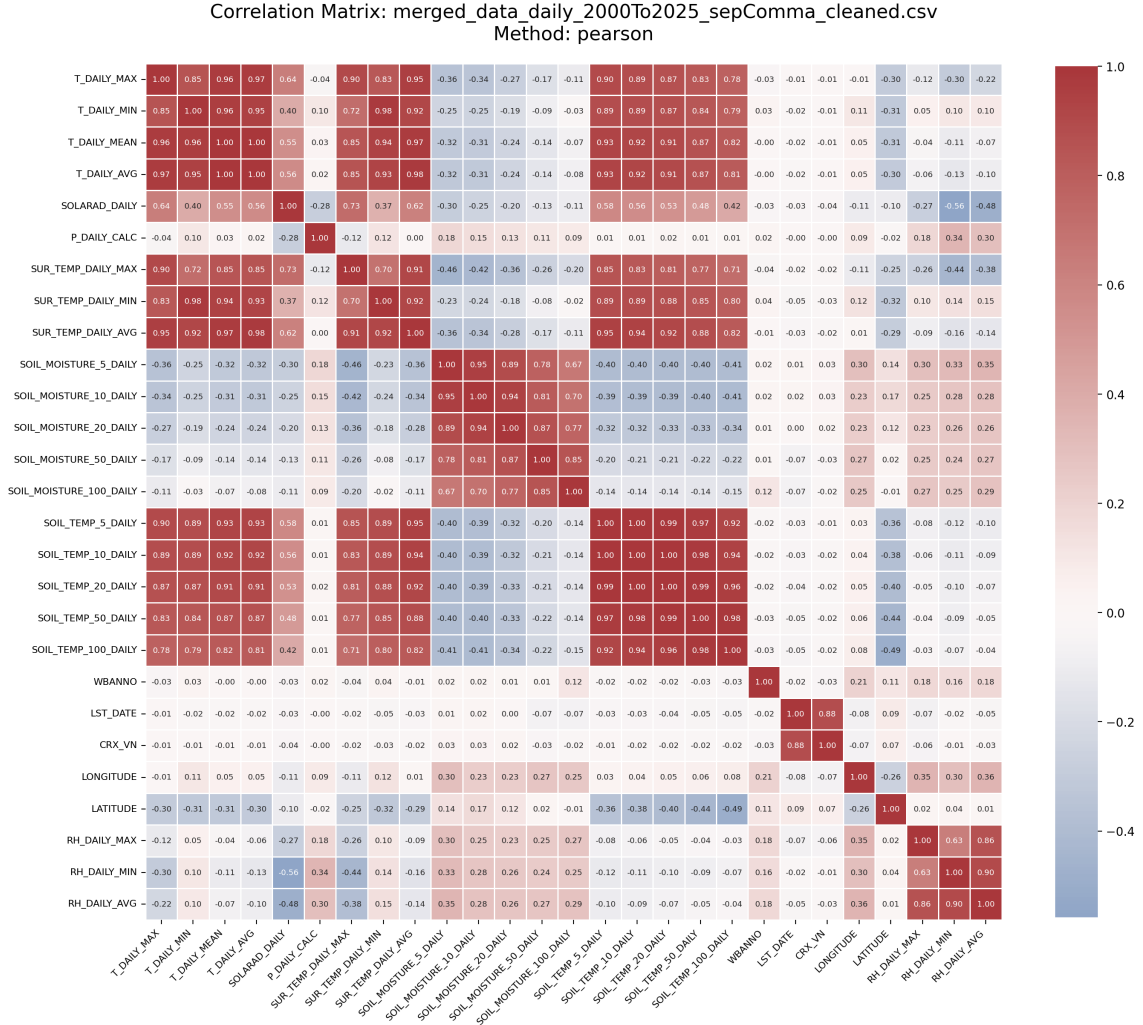


Figure 1: Correlation Matrix: Initial 23 Features

2.3 Spatial Visualization and Station Selection

To understand the geographic distribution of hydro-meteorological drivers, we generated a series of 10 nationwide maps representing key features across the United States. For several variables, **Min-Max Normalization** was applied to handle aberrant data points (outliers) and improve the visual clarity of the dispersion patterns.

2.3.1 Clustering of Homogeneous Stations

A critical innovation in our methodology is the transition from a "Global" model to a **Regional Homogeneous Model**. By plotting the mean and standard deviation (volatility) of moisture

Correlation Matrix: merged_data_daily_2000To2025_sepComma_cleaned.csv
Method: pearson

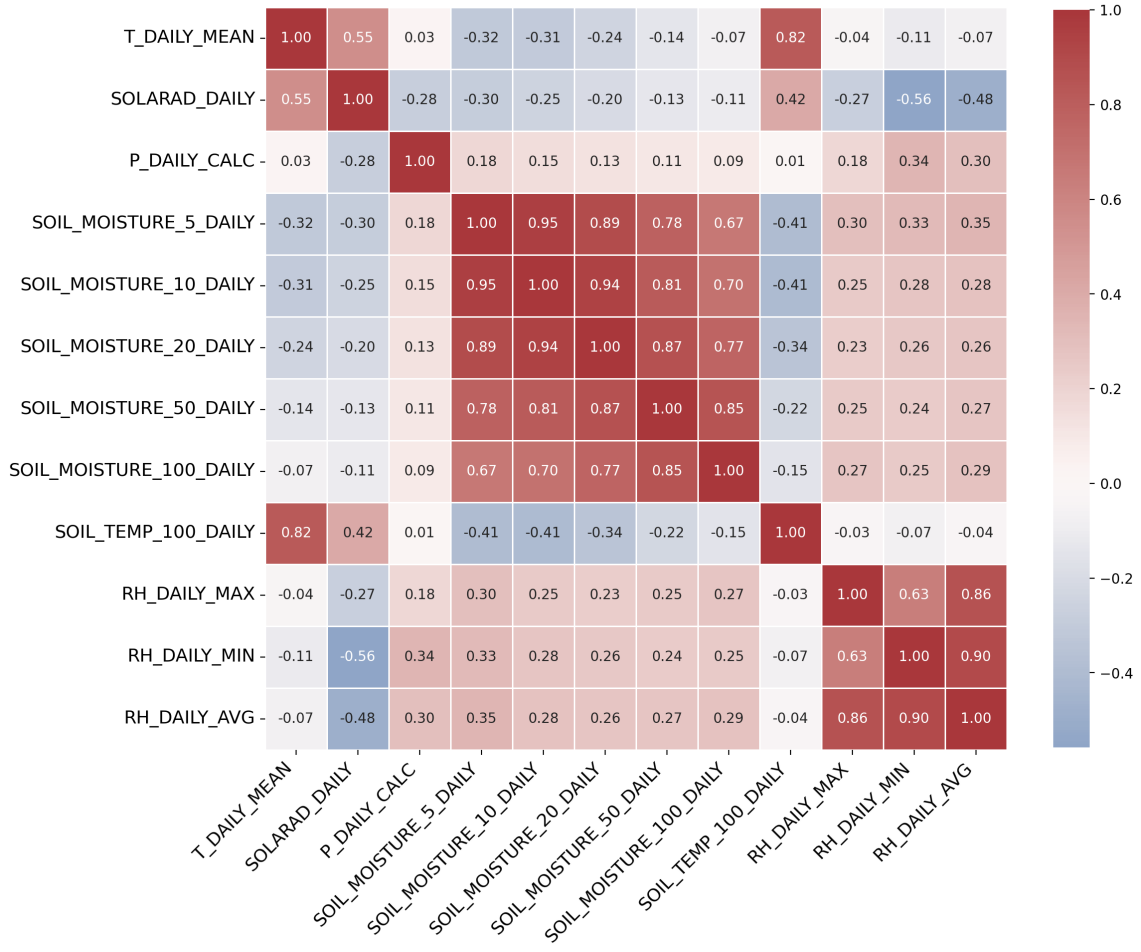


Figure 2: Correlation Matrix: Refined 12 Features

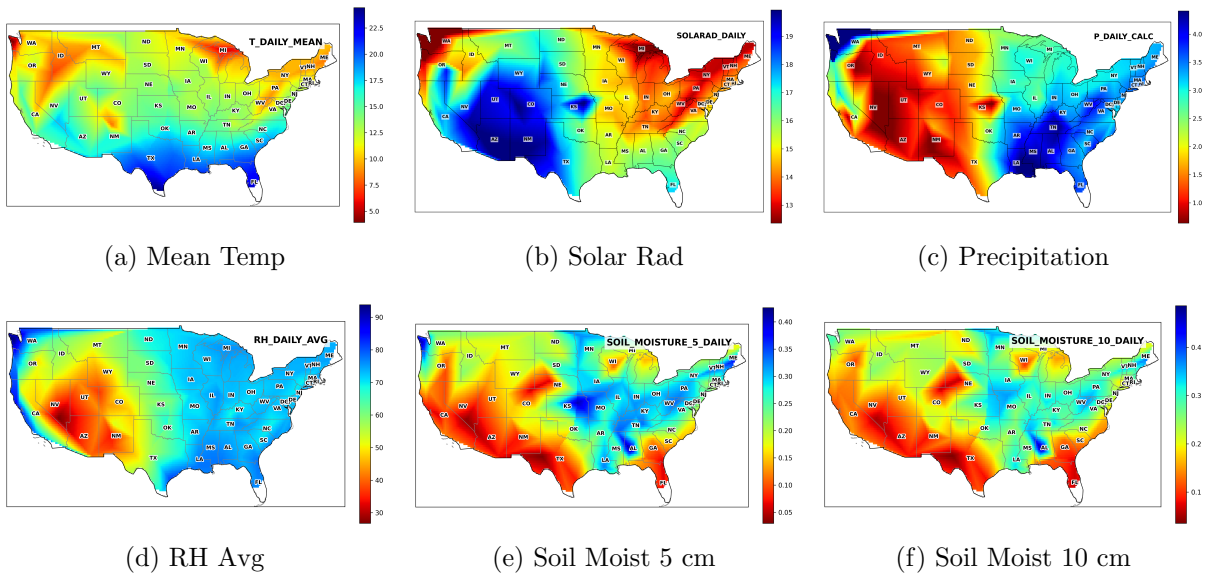


Figure 3: U.S. Geographic Distribution — Key meteorological and surface soil variables (6 maps, 3-per-row compact grid).

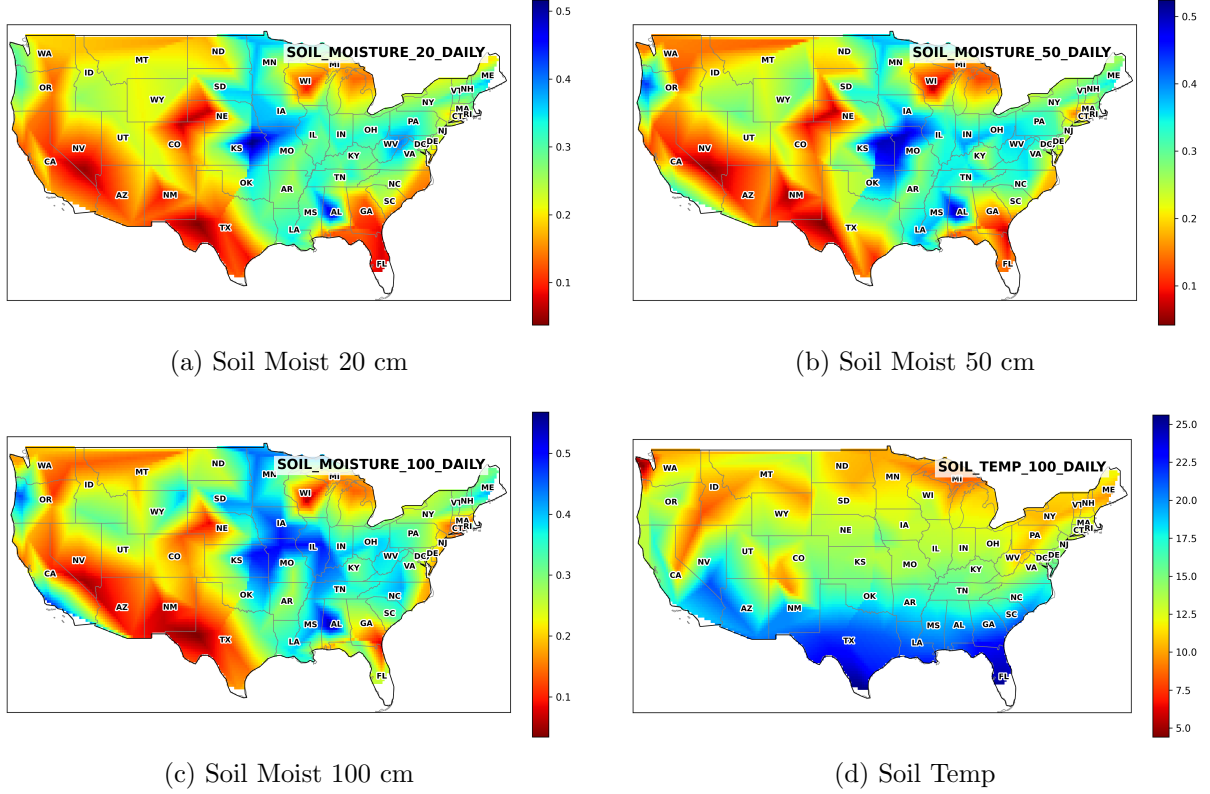


Figure 4: U.S. Geographic Distribution — Subsurface soil variables (4 maps, 2-per-row compact grid).

levels for every station, we identified a cluster of 5 stations exhibiting statistically similar behavior and low variance [17]. These stations (WBANNO IDs: 53877, 23904, 63857, 94996, and 53182) were used to create a targeted dataset of approximately 22,000 lines, ensuring the models learned from a consistent hydro-climatic signal rather than spatial noise [4].

2.4 Model Topology and Experimental Design

To determine the most effective architecture for one-day-ahead Volumetric Water Content (VWC) forecasting at a 5 cm depth, we implemented six distinct modeling approaches. Following the methodology of [3], we utilized a sliding window approach with a lookback sequence length of 10 days to predict the moisture state for the following 24-hour period.

2.4.1 Long Short-Term Memory (LSTM)

The LSTM model is designed to capture long-term temporal dependencies through a gating mechanism that regulates the flow of information [12, 20]. The state of the LSTM cell (c_t) and the hidden state (h_t) are updated as follows:

$$\begin{aligned}
 i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\
 f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\
 c_t &= f_t \circ c_{t-1} + i_t \circ \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \\
 o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \\
 h_t &= o_t \circ \tanh(c_t)
 \end{aligned} \tag{2}$$

Our architecture consists of a two-layer LSTM stack (128 and 64 units) with Dropout (0.2) to prevent overfitting [20]. This model processes 3D sequential data to learn the gradual drying

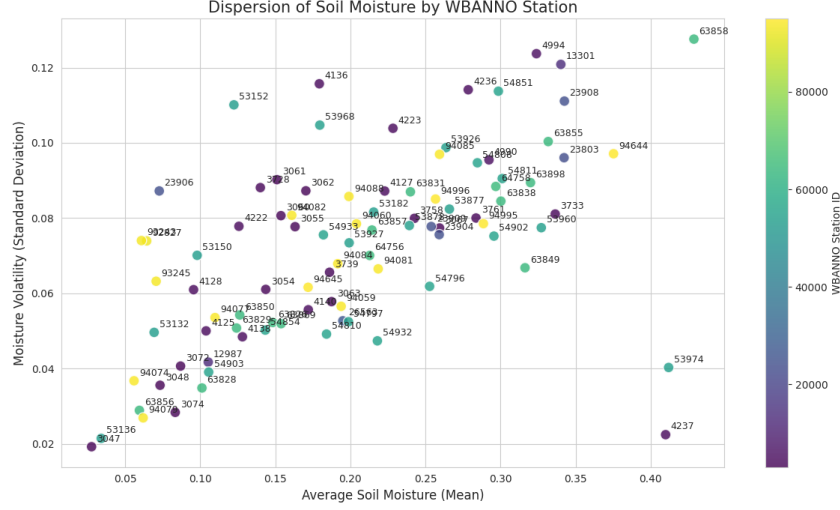


Figure 5: Dispersion of Soil Moisture by WBANNO Station: Identifying Homogeneous Clusters

and wetting cycles of the soil.

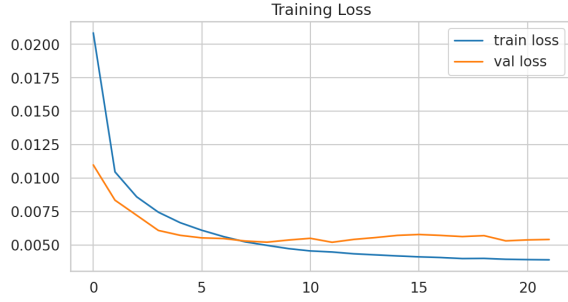


Figure 6: LSTM Training and Validation Loss (MSE) Convergence.

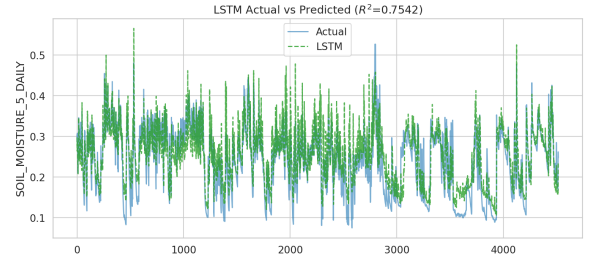


Figure 7: LSTM Performance: Actual vs. Predicted VWC (Test Set).

2.4.2 Extreme Gradient Boosting (XGBoost)

XGBoost is an ensemble learner that builds regression trees f_k sequentially to minimize the residual errors of the preceding trees [7, 16]. The final prediction $\hat{y}_i$ is the additive sum of K trees:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F} \quad (3)$$

Following the approach in [7], we utilized XGBoost to handle 48 lagged meteorological and soil features (lags 1, 3, 7, and 10), allowing the model to capture the "physical memory" of the soil system [7]. The XGBoost-based stacks provided the best one-day forecast accuracy in our regional experiments.

2.4.3 Standalone Transformer Architecture

The Transformer relies on the scaled dot-product attention mechanism to weigh the importance of different days in the 10-day lookback window [3, 13]. The attention function is defined as:

$$Attention(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (4)$$

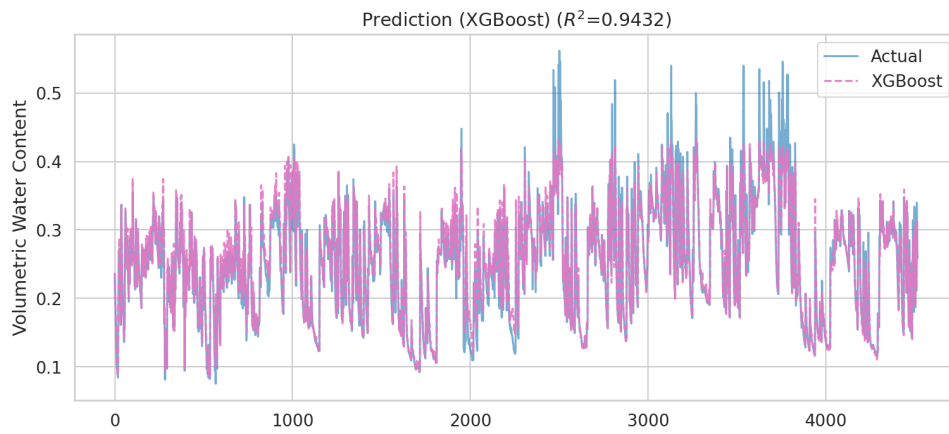


Figure 8: XGBoost Performance: Actual vs. Predicted VWC (Test Set).

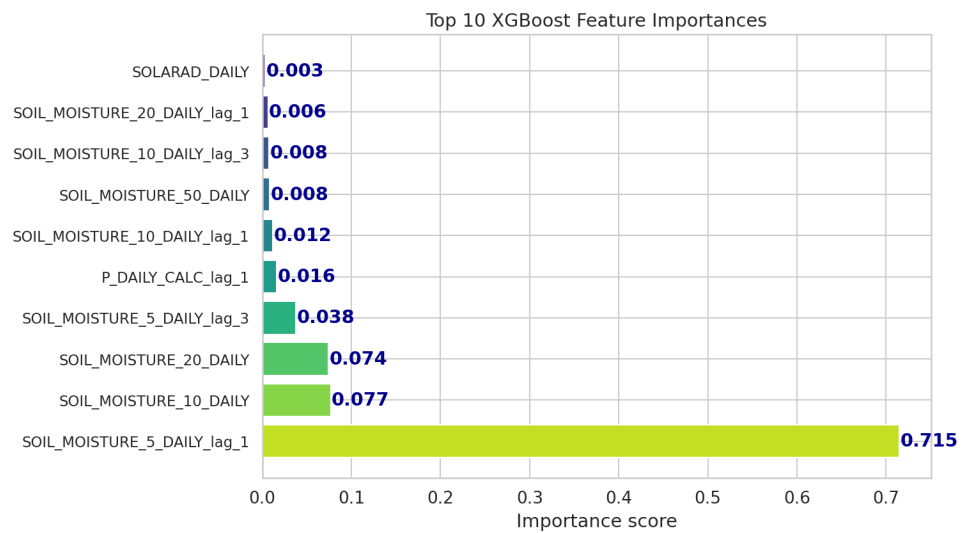


Figure 9: XGBoost Feature Importance: Top 10 Drivers of Soil Moisture.

By utilizing 4 attention heads and learned positional embeddings, the model can identify critical past events (such as intense precipitation 7 days prior) that influence tomorrow’s moisture level.



Figure 10: Transformer Training and Validation Loss (MSE) Convergence.

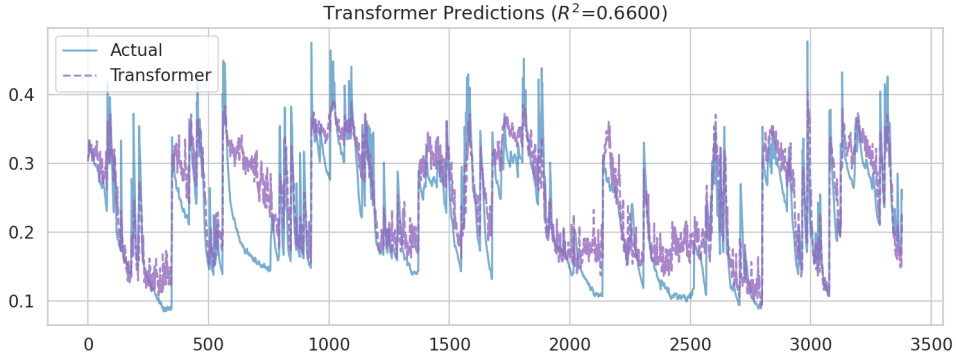


Figure 11: Transformer Performance: Sequential Prediction and Attention Weights.

2.4.4 Hybrid XGB+CNN Stack

This hybrid architecture uses a 1D Convolutional Neural Network (CNN) to extract local spatial-temporal features from meteorological sequences, while XGBoost handles the tabular lagged indicators. A meta-learner combines their outputs:

$$\hat{y}_{stack} = w_1(\text{XGB}) + w_2(\text{CNN}) + b \quad (5)$$

2.4.5 Hybrid XGB+Transformer Stack

The XGB+Transformer stack leverages the tree-based efficiency of gradient boosting alongside the global attention mechanism of the Transformer. This stacking topology is designed to correct the individual biases of each model, particularly during extreme wetting events where attention-based models often outperform standard trees.

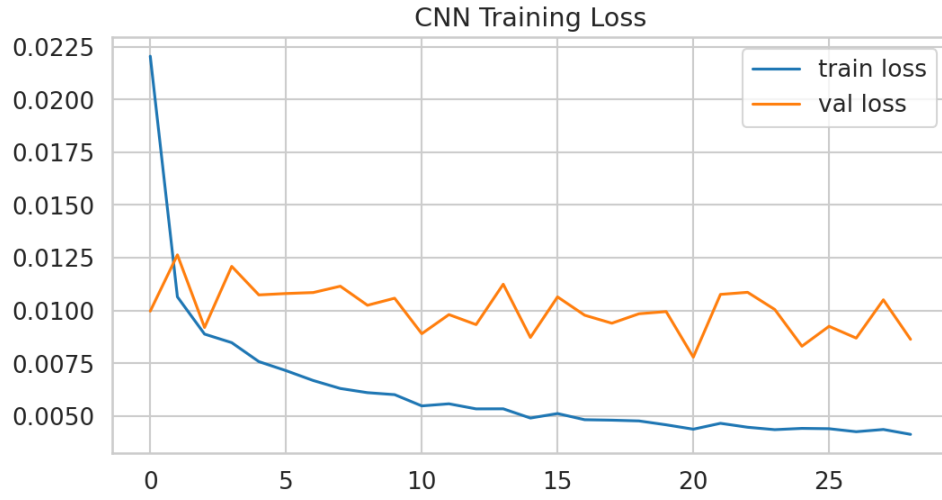


Figure 12: XGB+CNN Training and Validation Loss (MSE) Convergence.

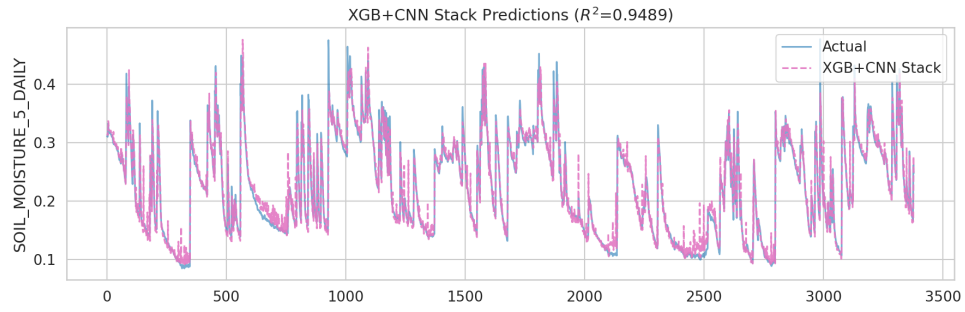


Figure 13: XGB+CNN Hybrid Stack: Combined Prediction Performance.

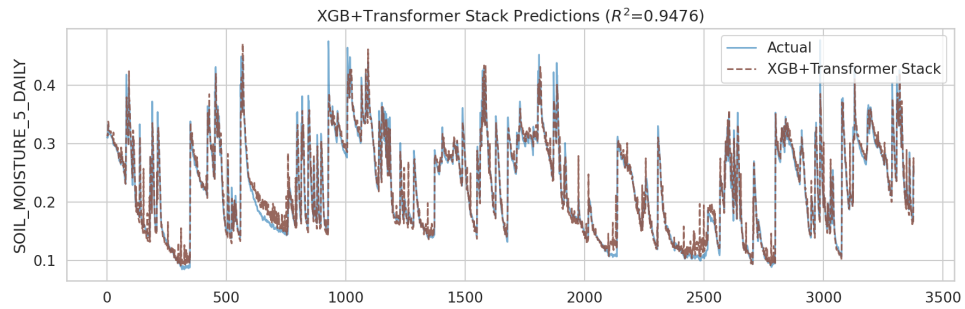


Figure 14: XGB+Transformer Hybrid Stack: Validation of Sequential-Tabular Fusion.

2.4.6 Hybrid XGB+GRU Stack

Gated Recurrent Units (GRU) are a simplified version of LSTM with fewer parameters (reset and update gates), making them computationally efficient. In this stack, the GRU captures temporal trends while XGBoost provides robustness against high-variance features in the meteorological dataset.

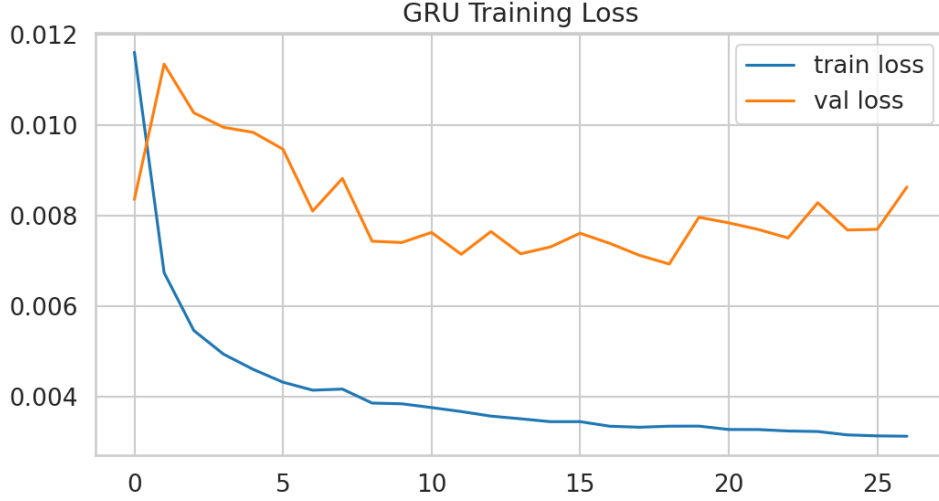


Figure 15: XGB+GRU Training and Validation Loss (MSE) Convergence.

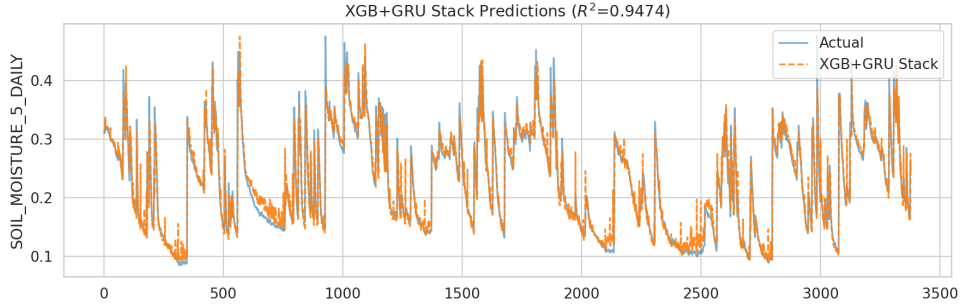


Figure 16: XGB+GRU Hybrid Stack: Performance Metrics and Temporal Fidelity.

3 Results and Discussion

3.1 Vertical Profile Correlation and Target Selection

A preliminary analysis was conducted to evaluate the relationship between Volumetric Water Content (VWC) at various depths. Consistent with the vertical heterogeneity discussed in [7], our correlation analysis (Fig. 2) revealed a strong statistical coupling between the surface layer (5 cm) and the deeper root zone (100 cm). Specifically, the correlation coefficient between adjacent depths (e.g., 5 cm to 10 cm) exceeded 0.95, and remained significant even at the 100 cm depth. This justification allows the 5 cm daily moisture level to serve as a robust representative for the entire profile, simplifying the modeling task without sacrificing regional accuracy.

3.2 Comparative Performance of Machine Learning Architectures

The performance of the six implemented architectures is summarized in Table 1. A critical observation from our initial experiments was the "persistence baseline" of 11%, where a simple

model predicted today’s moisture as being nearly identical to yesterday’s. By transitioning to a **Regional Homogeneous Model** targeting five similar stations, all advanced models successfully breached this ceiling, with hybrid architectures achieving accuracies exceeding 94%.

Table 1: Performance Metrics for One-Day-Ahead Soil Moisture Forecasting

Model Architecture	R^2 Score	MAE (m^3/m^3)	RMSE (m^3/m^3)
LSTM (Standalone)	0.7904	0.0268	0.0045
XGBoost (Standalone)	0.9432	0.0099	0.0261
Transformer (Standalone)	0.7138	0.0317	0.0082
XGB+CNN Stack	0.9491	0.0099	0.0084
XGB+Transformer Stack	0.9475	0.0105	0.0080
XGB+GRU Stack	0.9473	0.0105	0.0069

3.3 Discussion of Model Efficiency

3.3.1 Superiority of Hybrid Stacking

The results demonstrate that hybrid "stacked" models significantly outperform standalone deep learning architectures. While the standalone Transformer and LSTM models achieved R^2 values of 0.71 and 0.79 respectively, their combination with XGBoost pushed predictive accuracy to the 0.94-0.95 range. This performance leap is attributed to the meta-learner’s ability to combine XGBoost’s efficiency in handling tabular lagged features with the sequential feature extraction of neural networks. Specifically, the **XGB+CNN stack** achieved the highest overall score ($R^2 = 0.9491$), effectively capturing both the meteorological forcing and the local temporal trends.

3.3.2 The Persistence Problem and Regional Homogeneity

The jump from 11% to over 90% accuracy confirms that spatial heterogeneity is the primary limiting factor in national-scale soil moisture modeling. Global models struggle to find a universal rule for diverse climates, often defaulting to a simple persistence prediction. By clustering stations with statistically similar dispersion (as shown in the Station Dispersion Plot, Fig. 5), we reduced the signal-to-noise ratio, allowing the models to learn complex non-linear drivers such as solar radiation and antecedent precipitation.

3.4 Dominant Drivers of Soil Moisture

Feature importance analysis via XGBoost Gain (see Fig. 9) indicates that the most critical driver for tomorrow’s moisture level is the **one-day-lagged moisture content**. However, meteorological variables also played a vital role; Mean Temperature (T_DAILY_MEAN) and Solar Radiation ($SOLARAD_DAILY$) were identified as top drivers, reflecting the impact of atmospheric evaporative demand on soil drying cycles. These findings align with [7], confirming that RZSM dynamics are a function of both physical memory and recent atmospheric forcing.

3.5 Illustrative Model Output (referencing Methodology figures)

To avoid repeating the same images in multiple places in the manuscript, the model-specific actual vs predicted frames are presented where the model architectures are described (Section 2). Specifically:

- The XGB+CNN hybrid stack’s Actual vs Predicted VWC frame is shown in Fig. 13.
- The XGB+Transformer hybrid stack’s frame is shown in Fig. 14.

- The XGB+GRU hybrid stack’s frame is shown in Fig. 16.
- The standalone Transformer results are shown in Fig. 11.

This avoids duplicate images appearing in both Methodology and Results while keeping the figures close to their model descriptions for clarity.

4 Conclusion

This study investigated the efficacy of various machine learning and hybrid stacking architectures for one-day-ahead Volumetric Water Content (VWC) forecasting at a 5 cm depth. Our primary finding demonstrates that the inherent spatial heterogeneity in national-scale datasets is a significant barrier to predictive accuracy. While a global model trained on unclustered data yielded an R^2 of only 0.11, the implementation of a **Regional Homogeneous Modeling** strategy—clustering stations with statistically similar hydro-meteorological dispersion—pushed predictive accuracy to over 94%.

Among the six architectures evaluated, the **XGB+CNN hybrid stack** emerged as the superior model ($R^2 = 0.9491$), effectively leveraging the tabular efficiency of gradient-boosted trees alongside the local feature extraction capabilities of convolutional neural networks. The study also confirmed a strong vertical coupling across the soil profile, justifying the use of surface moisture as a reliable proxy for deeper root-zone states.

The high fidelity of these models in predicting a one-day lag provides a robust foundation for precision agriculture. By accurately forecasting tomorrow’s moisture levels, this framework offers a data-driven tool for optimized irrigation scheduling, potentially reducing water waste and enhancing crop resilience. Future research should explore the integration of dynamic vegetation indices and high-resolution soil texture data to further improve model interpretability and generalizability across diverse climatic zones.

Webography

1. **NOAA National Centers for Environmental Information:** U.S. Climate Reference Network (USCRN) high-resolution daily meteorological and soil moisture station data.
URL: <https://www.ncei.noaa.gov/>
2. **ESA CCI Soil Moisture:** European Space Agency Climate Change Initiative global daily soil moisture products (Surface and Root Zone).
URL: <https://www.esa-soilmoisture-cci.org/>
3. **International Soil Moisture Network (ISMN):** Global repository for in situ soil moisture measurements and validation data.
URL: <https://ismn.geo.tuwien.ac.at/>

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